

**MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY,
BHOPAL - 462003**

Name of Program: B.Tech (Electrical)		Semester: IV	Year: 2 nd
Name of Course:		Generation of Electrical Power	
Course Code:		EE 224	
Core/Elective/other:		Core	
Prerequisite: Fundamentals of Electrical Machines, Prime movers, Engineering Mathematics			
Course Outcome: After studying this subject, students will be able to			
01	Understand structure of modern power systems		
02	explain the basic principle of Generation of Electrical power from various sources		
03	analyze economics of power plants		
Description of Contents in brief			
1	Introduction to electrical power generation and classification of conventional & nonconventional power generation, principle of operation of solar, wind and tidal power, cogeneration, load curves, performance factors		
2	tariffs, basic structures of electricity markets, power factor improvement by static and synchronous capacitors, concept of reactive power compensation, concept of demand side management.		
3	thermal Power Stations: Choice of coal fired station site, arrangement of plant and principal auxiliaries, coal handling plant,		
4	Nuclear Power Stations, nuclear physics, site selection of nuclear stations, classification of reactors, working principle of reactors.		
5	Hydro-Electric Stations, site selection of hydro stations, hydrology, mass curve, flow duration curve, water storage, classification, main components, pumped storage plants.		
6	Economic Aspects of Power Plant Operation Economic Scheduling of Power Stations, Hydro-thermal coordination.		
List of Text Books			
01	B.R. Gupta, Generation of Electrical Energy, S Chand publisher		
02	M.V.Deshpande, Elements of power station design, Prentice Hall India Learning Private Limited.		
03	G.D. Rai, Non-conventional Energy Sources, Khanna Publishers		
List of Reference Books			
01	G.R. Nagpal, Power Plant Engineering, 15 th Edition, Khanna Publishers		
02	L. K. Kirchmayer, Economic operation of power systems, Wiley Eastern Ltd.		
URLs			
1.	http://nptel.ac.in (for “Electrical Engineering Energy Resources & Technologies”)		